

1 Linear Combinations

$$i^3 b_{2u} * i^3 b_{2u}$$

$$+ \left(\begin{array}{cccc} \uparrow & - & \uparrow & - \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{array} \right) - \left(\begin{array}{cccc} \uparrow & - & - & \uparrow \\ \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow \end{array} \right) - \left(\begin{array}{cccc} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow \end{array} \right) + \left(\begin{array}{cccc} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow\uparrow & \uparrow & \downarrow\uparrow \end{array} \right)$$

written in CI vectors:

monomer ci vectors:

$$+ |(b_{1u}^r)^1 (b_{2g}^r)^2 (b_{3g}^r)^1 (a_u^r)^0| \cdot |(b_{1u}^r)^1 (b_{2g}^r)^2 (b_{3g}^r)^1 (a_u^r)^0| - |(b_{1u}^r)^1 (b_{2g}^r)^2 (b_{3g}^r)^1 (a_u^r)^0| \cdot |(b_{1u}^r)^0 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^1| - |(b_{1u}^r)^0 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^1| \cdot |(b_{1u}^r)^1 (b_{2g}^r)^2 (b_{3g}^r)^1 (a_u^r)^0|$$

substitution by dimer:

$$\begin{aligned} & + |(-a_g + b_{1u})^1 (-b_{3u} + b_{2g})^2 (-b_{2u} + b_{3g})^1 (-b_{1g} + a_u)^0| \cdot |(-a_g + b_{1u})^1 (-b_{3u} + b_{2g})^2 (-b_{2u} + b_{3g})^1 (-b_{1g} + a_u)^0| \\ & - |(-a_g + b_{1u})^1 (-b_{3u} + b_{2g})^2 (-b_{2u} + b_{3g})^1 (-b_{1g} + a_u)^0| \cdot |(-a_g + b_{1u})^0 (-b_{3u} + b_{2g})^1 (-b_{2u} + b_{3g})^2 (-b_{1g} + a_u)^1| \\ & - |(-a_g + b_{1u})^0 (-b_{3u} + b_{2g})^1 (-b_{2u} + b_{3g})^2 (-b_{1g} + a_u)^1| \cdot |(-a_g + b_{1u})^1 (-b_{3u} + b_{2g})^2 (-b_{2u} + b_{3g})^1 (-b_{1g} + a_u)^0| \\ & + |(-a_g + b_{1u})^0 (-b_{3u} + b_{2g})^1 (-b_{2u} + b_{3g})^2 (-b_{1g} + a_u)^1| \cdot |(-a_g + b_{1u})^0 (-b_{3u} + b_{2g})^1 (-b_{2u} + b_{3g})^2 (-b_{1g} + a_u)^1| \end{aligned}$$

dimer ci vectors:

$$\begin{aligned} & + |(a_g)^1 (b_{3u})^2 (b_{2u})^1 (b_{1g})^0 (b_{1u})^1 (b_{2g})^2 (b_{3g})^1 (a_u)^0| - |(a_g)^1 (b_{3u})^2 (b_{2u})^1 (b_{1g})^0 (b_{1u})^0 (b_{2g})^1 (b_{3g})^2 (a_u)^1| \\ & - |(a_g)^1 (b_{3u})^2 (b_{2u})^1 (b_{1g})^1 (b_{1u})^0 (b_{2g})^1 (b_{3g})^2 (a_u)^0| - |(a_g)^1 (b_{3u})^2 (b_{2u})^2 (b_{1g})^0 (b_{1u})^0 (b_{2g})^1 (b_{3g})^1 (a_u)^1| \\ & - |(a_g)^1 (b_{3u})^2 (b_{2u})^2 (b_{1g})^1 (b_{1u})^0 (b_{2g})^1 (b_{3g})^1 (a_u)^0| - |(a_g)^1 (b_{3u})^1 (b_{2u})^1 (b_{1g})^0 (b_{1u})^0 (b_{2g})^2 (b_{3g})^2 (a_u)^1| \\ & - |(a_g)^1 (b_{3u})^1 (b_{2u})^1 (b_{1g})^1 (b_{1u})^0 (b_{2g})^2 (b_{3g})^2 (a_u)^0| - |(a_g)^1 (b_{3u})^1 (b_{2u})^2 (b_{1g})^0 (b_{1u})^0 (b_{2g})^2 (b_{3g})^1 (a_u)^1| \\ & - |(a_g)^1 (b_{3u})^1 (b_{2u})^2 (b_{1g})^1 (b_{1u})^0 (b_{2g})^2 (b_{3g})^1 (a_u)^0| - |(a_g)^0 (b_{3u})^2 (b_{2u})^1 (b_{1g})^0 (b_{1u})^1 (b_{2g})^1 (b_{3g})^2 (a_u)^1| \\ & - |(a_g)^0 (b_{3u})^2 (b_{2u})^1 (b_{1g})^1 (b_{1u})^1 (b_{2g})^1 (b_{3g})^2 (a_u)^0| - |(a_g)^0 (b_{3u})^2 (b_{2u})^2 (b_{1g})^0 (b_{1u})^1 (b_{2g})^1 (b_{3g})^1 (a_u)^1| \\ & - |(a_g)^0 (b_{3u})^2 (b_{2u})^2 (b_{1g})^1 (b_{1u})^1 (b_{2g})^1 (b_{3g})^1 (a_u)^0| - |(a_g)^0 (b_{3u})^1 (b_{2u})^1 (b_{1g})^0 (b_{1u})^1 (b_{2g})^2 (b_{3g})^2 (a_u)^1| \\ & - |(a_g)^0 (b_{3u})^1 (b_{2u})^1 (b_{1g})^1 (b_{1u})^1 (b_{2g})^2 (b_{3g})^2 (a_u)^0| - |(a_g)^0 (b_{3u})^1 (b_{2u})^2 (b_{1g})^0 (b_{1u})^1 (b_{2g})^2 (b_{3g})^1 (a_u)^1| \\ & - |(a_g)^0 (b_{3u})^1 (b_{2u})^2 (b_{1g})^1 (b_{1u})^1 (b_{2g})^2 (b_{3g})^1 (a_u)^0| - |(a_g)^0 (b_{3u})^1 (b_{2u})^2 (b_{1g})^1 (b_{1u})^1 (b_{2g})^2 (b_{3g})^1 (a_u)^0| \\ & - |(a_g)^0 (b_{3u})^1 (b_{2u})^2 (b_{1g})^0 (b_{1u})^1 (b_{2g})^2 (b_{3g})^1 (a_u)^1| - |(a_g)^0 (b_{3u})^1 (b_{2u})^1 (b_{1g})^1 (b_{1u})^1 (b_{2g})^2 (b_{3g})^2 (a_u)^0| \\ & - |(a_g)^0 (b_{3u})^1 (b_{2u})^1 (b_{1g})^0 (b_{1u})^1 (b_{2g})^2 (b_{3g})^2 (a_u)^1| - |(a_g)^0 (b_{3u})^2 (b_{2u})^2 (b_{1g})^1 (b_{1u})^1 (b_{2g})^1 (b_{3g})^1 (a_u)^0| \\ & - |(a_g)^0 (b_{3u})^2 (b_{2u})^2 (b_{1g})^0 (b_{1u})^1 (b_{2g})^1 (b_{3g})^1 (a_u)^1| - |(a_g)^0 (b_{3u})^2 (b_{2u})^1 (b_{1g})^1 (b_{1u})^0 (b_{2g})^2 (b_{3g})^1 (a_u)^0| \\ & - |(a_g)^1 (b_{3u})^1 (b_{2u})^2 (b_{1g})^0 (b_{1u})^0 (b_{2g})^2 (b_{3g})^1 (a_u)^1| - |(a_g)^1 (b_{3u})^1 (b_{2u})^1 (b_{1g})^1 (b_{1u})^0 (b_{2g})^2 (b_{3g})^2 (a_u)^0| \\ & - |(a_g)^1 (b_{3u})^1 (b_{2u})^1 (b_{1g})^0 (b_{1u})^0 (b_{2g})^2 (b_{3g})^2 (a_u)^1| - |(a_g)^1 (b_{3u})^2 (b_{2u})^2 (b_{1g})^1 (b_{1u})^0 (b_{2g})^1 (b_{3g})^1 (a_u)^0| \\ & - |(a_g)^1 (b_{3u})^2 (b_{2u})^2 (b_{1g})^0 (b_{1u})^0 (b_{2g})^1 (b_{3g})^1 (a_u)^1| - |(a_g)^1 (b_{3u})^2 (b_{2u})^1 (b_{1g})^1 (b_{1u})^0 (b_{2g})^1 (b_{3g})^2 (a_u)^0| \\ & - |(a_g)^1 (b_{3u})^2 (b_{2u})^1 (b_{1g})^0 (b_{1u})^0 (b_{2g})^1 (b_{3g})^2 (a_u)^1| + |(a_g)^0 (b_{3u})^1 (b_{2u})^2 (b_{1g})^1 (b_{1u})^0 (b_{2g})^1 (b_{3g})^2 (a_u)^1| \end{aligned}$$

summed up:

$$\begin{aligned}
& + |(a_g)^1(b_{3u})^2(b_{2u})^1(b_{1g})^0(b_{1u})^1(b_{2g})^2(b_{3g})^1(a_u)^0| - 2 \cdot |(a_g)^1(b_{3u})^2(b_{2u})^1(b_{1g})^0(b_{1u})^0(b_{2g})^1(b_{3g})^2(a_u)^1| \\
& - 2 \cdot |(a_g)^1(b_{3u})^2(b_{2u})^1(b_{1g})^1(b_{1u})^0(b_{2g})^1(b_{3g})^2(a_u)^0| - 2 \cdot |(a_g)^1(b_{3u})^2(b_{2u})^2(b_{1g})^0(b_{1u})^0(b_{2g})^1(b_{3g})^1(a_u)^1| \\
& - 2 \cdot |(a_g)^1(b_{3u})^2(b_{2u})^2(b_{1g})^1(b_{1u})^0(b_{2g})^1(b_{3g})^1(a_u)^0| - 2 \cdot |(a_g)^1(b_{3u})^1(b_{2u})^1(b_{1g})^0(b_{1u})^0(b_{2g})^2(b_{3g})^2(a_u)^1| \\
& - 2 \cdot |(a_g)^1(b_{3u})^1(b_{2u})^1(b_{1g})^1(b_{1u})^0(b_{2g})^2(b_{3g})^2(a_u)^0| - 2 \cdot |(a_g)^1(b_{3u})^1(b_{2u})^2(b_{1g})^0(b_{1u})^0(b_{2g})^2(b_{3g})^1(a_u)^1| \\
& - 2 \cdot |(a_g)^1(b_{3u})^1(b_{2u})^2(b_{1g})^1(b_{1u})^0(b_{2g})^2(b_{3g})^1(a_u)^0| - 2 \cdot |(a_g)^0(b_{3u})^2(b_{2u})^1(b_{1g})^0(b_{1u})^1(b_{2g})^1(b_{3g})^2(a_u)^1| \\
& - 2 \cdot |(a_g)^0(b_{3u})^2(b_{2u})^1(b_{1g})^1(b_{1u})^1(b_{2g})^1(b_{3g})^2(a_u)^0| - 2 \cdot |(a_g)^0(b_{3u})^2(b_{2u})^2(b_{1g})^0(b_{1u})^1(b_{2g})^1(b_{3g})^1(a_u)^1| \\
& - 2 \cdot |(a_g)^0(b_{3u})^2(b_{2u})^2(b_{1g})^1(b_{1u})^1(b_{2g})^1(b_{3g})^1(a_u)^0| - 2 \cdot |(a_g)^0(b_{3u})^1(b_{2u})^1(b_{1g})^0(b_{1u})^1(b_{2g})^2(b_{3g})^2(a_u)^1| \\
& - 2 \cdot |(a_g)^0(b_{3u})^1(b_{2u})^1(b_{1g})^1(b_{1u})^1(b_{2g})^2(b_{3g})^2(a_u)^0| - 2 \cdot |(a_g)^0(b_{3u})^1(b_{2u})^2(b_{1g})^0(b_{1u})^1(b_{2g})^2(b_{3g})^1(a_u)^1| \\
& - 2 \cdot |(a_g)^0(b_{3u})^1(b_{2u})^2(b_{1g})^1(b_{1u})^1(b_{2g})^2(b_{3g})^1(a_u)^0| + |(a_g)^0(b_{3u})^1(b_{2u})^2(b_{1g})^1(b_{1u})^0(b_{2g})^1(b_{3g})^2(a_u)^1|
\end{aligned}$$

or, expressed in terms of symmetry:

monomer determinants:

$$+1 \cdot |(b_{1u}^l)(b_{3g}^l)(b_{1u}^r)(b_{3g}^r)| - 1 \cdot |(b_{1u}^l)(b_{3g}^l)(b_{2g}^r)(a_u^r)| - 1 \cdot |(b_{2g}^l)(a_u^l)(b_{1u}^r)(b_{3g}^r)| + 1 \cdot |(b_{2g}^l)(a_u^l)(b_{2g}^r)(a_u^r)|$$

substitution by dimer:

$$\begin{aligned}
& +1 \cdot |(a_g + b_{1u})(a_g + b_{2u} + b_{3g})(-a_g + b_{1u})(-b_{2u} + b_{3g})| - 1 \cdot |(a_g + b_{1u})(a_g + b_{2u} + b_{3g})(-b_{3u} + b_{2g})(-b_{1g} + a_u)| \\
& -1 \cdot |(a_g + b_{1u})(a_g + b_{2u} + b_{3g})(-a_g + b_{1u})(-b_{2u} + b_{3g})| + 1 \cdot |(a_g + b_{1u})(a_g + b_{2u} + b_{3g})(-b_{3u} + b_{2g})(-b_{1g} + a_u)|
\end{aligned}$$

multiplied out:

$$\begin{aligned}
& + \underbrace{a_g b_{2u} b_{1u} b_{3g}}_{a_g} - \underbrace{a_g b_{2u} b_{2g} a_u}_{a_g} - \underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}} - \underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}} - \underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g} - \underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}} \\
& - \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} - \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} - \underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}} - \underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}} - \underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g} - \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g} \\
& - \underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}} - \underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g} - \underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}} - \underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}} - \underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g} - \underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g} \\
& - \underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}} - \underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}} - \underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g} - \underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}} - \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g} - \underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g} \\
& - \underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}} - \underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}} - \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} - \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} - \underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}} - \underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g} \\
& - \underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}} - \underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}} - \underbrace{a_g b_{2u} b_{2g} a_u}_{a_g} + \underbrace{b_{3u} b_{1g} b_{2g} a_u}_{a_g}
\end{aligned}$$

summed up:

$$\begin{aligned}
& + \left| \underbrace{a_g b_{2u} b_{1u} b_{3g}}_{a_g} \right| - 2 \cdot \left| \underbrace{a_g b_{2u} b_{2g} a_u}_{a_g} \right| - 2 \cdot \left| \underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}} \right| - 2 \cdot \left| \underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}} \right| - 2 \cdot \left| \underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g} \right| - 2 \cdot \left| \underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}} \right| \\
& - 2 \cdot \left| \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} \right| - 2 \cdot \left| \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} \right| - 2 \cdot \left| \underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}} \right| - 2 \cdot \left| \underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}} \right| - 2 \cdot \left| \underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g} \right| - 2 \cdot \left| \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g} \right| \\
& - 2 \cdot \left| \underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}} \right| - 2 \cdot \left| \underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g} \right| - 2 \cdot \left| \underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}} \right| - 2 \cdot \left| \underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}} \right| - 2 \cdot \left| \underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g} \right| + \left| \underbrace{b_{3u} b_{1g} b_{2g} a_u}_{a_g} \right|
\end{aligned}$$
